



ENVIRONMENTAL SCIENCE

In Ukraine, dam's destruction sets off a 'toxic time bomb'

Floods threaten to spread sediments laden with toxicants

By Richard Stone

In the Soviet Union, the Zaporozhets automobile was legend. Mass produced in the 1960s in the Ukrainian city of Zaporizhzhia, early models were known for steel frames as flimsy as cardboard and a gas tank under the front hood that could turn a fender bender into a barbecue. But the plebeian car did have a classy touch: nickel-plated bumpers, which had consequences that reverberate today. For decades, factories making Zaporozhets components, along with other industry and agriculture, poured effluents laden with nickel, cadmium, lead, and other heavy metals into the huge Kakhovka Reservoir nearby, where the toxicants settled into the lake-bottom sludge.

On 6 June 2023, those sediments triggered what Alexander Sukhodolov, a hydrodynamicist at the Leibniz Institute of Freshwater Ecology and Inland Fisheries, calls “a toxic time bomb.” A 400-meter-wide section of the Kakhovka Dam collapsed, perhaps as a result of sabotage, sending 16.4 cubic kilometers of water—and tons of contaminated silt—surging down the Dnipro River in southern Ukraine. Scores of people died in the flood, up to 1 million lost access to drinking water, and irrigation canals for an important agricultural region turned into trickles. The deluge laid waste to ecosystems, leaving billions of mussels rotting in the desiccated lakebed and scrambling

estuaries where the river empties into the Black Sea (*Science*, 5 January 2024, p. 18).

In the months since the disaster, “ecosystems have shown a remarkable resiliency,” says Yuriy Kvach, a biologist at the National Academy of Sciences of Ukraine’s Institute of Marine Biology. Habitats and species are reviving, he and colleagues reported in *Ecological Processes* in February. Now that the Dnipro is flowing freely, endangered sturgeon have reappeared, and meadows dotted with willows are taking hold on the dry lakebed.

But behind the vibrancy lurks a potent pollution threat, Sukhodolov and colleagues detail in a paper this week in *Science* (p. 1181). Before the dam breach, Kakhovka’s reservoir accumulated a layer of loess silt up to 1.5 meters thick, amounting to as much as 1.7 cubic kilometers. The now exposed sediments contain about 83,000 tons of heavy metals, the researchers estimate. YouTube videos showing a brownish black crust on the lakebed are evidence of “the really huge accumulations of heavy metals,” says ecologist Oleksandra Shumilova, lead author on the *Science* paper who’s also at the Leibniz Institute.

The emerging danger is that those toxicants won’t stay put, Shumilova says. Less than 1% of the sediment has been swept downstream so far, but seasonal floods from heavy rains or snowmelts will continue to wash the pollutants down the Dnipro and into the water-

“No swimming! Careful mines!” warns a sign posted in the Kakhovka Reservoir’s desiccated bed.

shed around the former reservoir. In spring 2024, for instance, contaminated floodwaters inundated nearly 900 square kilometers, the *Science* authors report.

The war in Ukraine has stymied efforts to understand the scope of the threat. Much of the lower Dnipro winds along the front line, leaving scientists who venture there vulnerable to shelling or drone attacks. And the Ukrainian military forbids research cruises in the Black Sea, forcing scientists to rely mainly on remote sensing data to monitor water conditions. But some upstream areas, near Zaporizhzhia city, are safer to access. In 2023, Czech and Ukrainian nongovernmental organizations sampled the river at several sites, finding a witch’s brew including heavy metals, polyaromatic hydrocarbons, and the insecticide DDT.

The “real danger” lies in the pollutants building up, or bioaccumulating, in the food web, Shumilova says. So far, however, scientists have little data on that. To fill the gap, Shumilova and Ukrainian colleagues are meeting next month to develop a research plan for tracking bioaccumulation, for instance by probing for heavy metals in the scat of local deer populations.

In the meantime, researchers and others are debating whether the Kakhovka Dam should be rebuilt. Some ecologists think that’s a bad idea, noting that the willow meadows now thriving in the former lakebed are stabilizing sediments and sopping up heavy metals from the soil. Others argue rebuilding is necessary to restore irrigation networks and ensure drinking water supplies.

Sukhodolov cautions that the willows’ powers of remediation won’t eliminate the threat of heavy metals borne on seasonal floods. He expects Ukraine to rebuild the dam, and until the river is impounded again he and his co-authors propose building two 15-kilometer-long barriers along the Dnipro to curtail pollution releases.

As long as the war continues, such plans are pipe dreams. In the meantime, the specter of another threat haunts Ukraine: possible attacks on the Dnipro’s five remaining dams or on the Dniester River cascade, a series of hydropower and flood control structures. “If more dams are targeted, the human toll and environmental damage could be cataclysmic,” the *Science* authors warn.

“Water has been a weapon of war throughout history,” says ecologist Carol Stepien of the Smithsonian Institution’s National Museum of Natural History, who has worked in the Lower Dnipro Basin. Until the war ends, she says, “Ukraine will be in a rather precarious situation.” ■